

The Era of Big Data: Databases, Information Systems, & Artificial Intelligence

Chapter 8



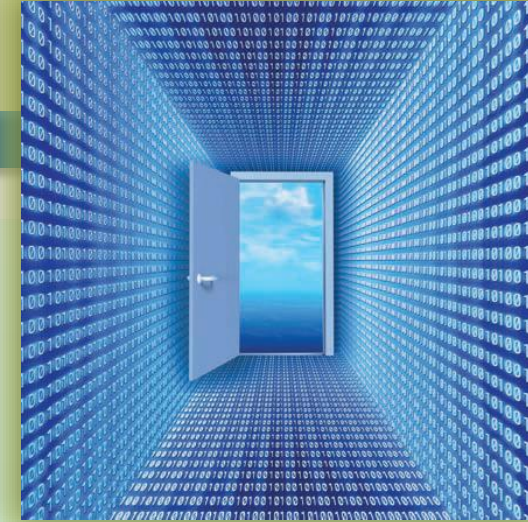
Chapter Topics

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- 8.1 Managing Files: Basic Concepts
- 8.2 Database Management Systems
- 8.3 Database Models
- 8.4 Data Mining

UNIT 8B: Big Data, Information Systems, & Artificial Intelligence

- 8.5 The Evolving World of Big Data
- 8.6 Information Systems in Organizations: Using Databases to Help Make Decisions
- 8.7 Artificial Intelligence
- 8.8 Artificial Life, the Turing Test, & the Singularity



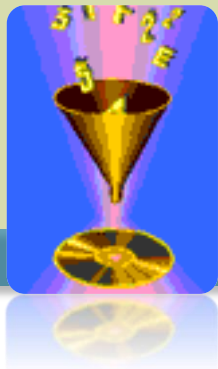


UNIT 8A: Files & Databases

- **Big Data** is so large and complex that it cannot be processed using conventional methods, such as ordinary database management software.
- Some experts expect data to grow by 20 times between 2012 and 2020.

8.1 Managing Files

Basic Concepts



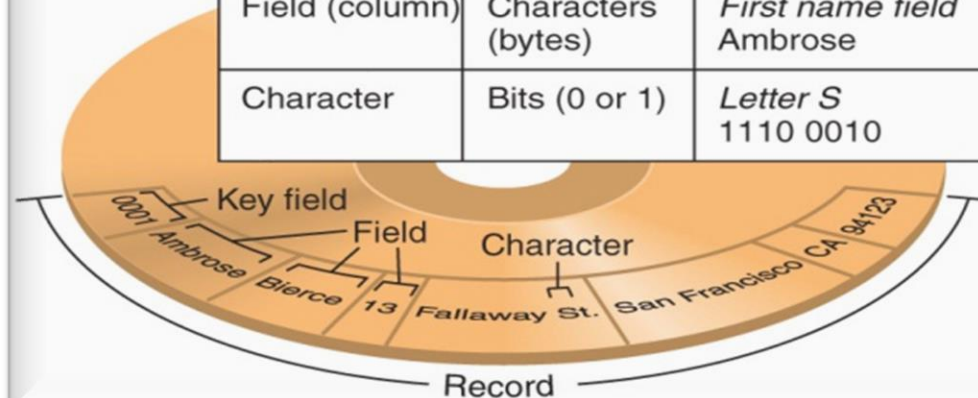
- A **database** is a logically organized collection of related data designed and built for a specific purpose.
- Data is stored **hierarchically** for easier storage and retrieval.
- **File (table)**: collection of related records
 - **Records (row)**: collections of related fields
 - **Field (column)**: unit of data containing 1 or more characters
 - **Character [Byte]**: a letter number or special character made of bits
 - **Bit**: 0 or 1



Data Storage Hierarchy



Type of data	Contains	Example
Database	Several files	<i>Your personal database</i> Friends' addresses file, CD titles file, Term papers file, etc.
Table (file)	Several records	<i>Friends' addresses file</i> Bierce, Ambrose 0001; London, Jack 0234; Stevenson, Robert L. 0081; etc.
Record (row)	Several fields	<i>Ambrose Bierce's name and address</i> 13 Fallaway St. San Francisco, CA 94123
Field (column)	Characters (bytes)	<i>First name field</i> Ambrose
Character	Bits (0 or 1)	<i>Letter S</i> 1110 0010





A **key field** (primary key) is a field (or fields) in a record that holds unique data that identifies that record from all the other records in the table and in the database.

- Often an identifying number, such as social security number or a student ID number.
- Keys are used to sort records in different ways.
- Primary keys must be unique make records distinguishable from one another.
- Foreign keys appear in other tables and usually refer to primary keys in particular tables; they are used to relate one table to another (to cross-reference data).

8.2 Database Management Systems



Database Management System (DBMS): software that enables users to store, modify, and extract information from a database

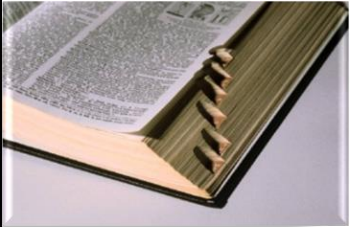
- DBMS benefits:
 - **Reduced data redundancy** (redundant data is stored in multiple places, which causes problems keeping all the copies current)
 - **Speed**—Modern DBMSs are much faster than manual data-organization systems and faster than older computer-based database arrangements
 - **Improved data integrity**—the data is accurate, consistent, and up to date
 - **Timeliness**—The speed and efficiency of DBMSs generally ensure that data can be supplied in a timely fashion—when people need it
 - **Ease of sharing**—The data in a database belongs to and is shared, usually over a network, by an entire organization. The data is independent of the programs that process the data, and it is easy for nontechnical users to access it.



- **Ease of data maintenance**—DBMS offers validation checks, backup utilities, and standard procedures for data inserting, updating, and deletion
- **Forecasting capabilities**—DBMSs can hold massive amounts of data that can be manipulated, studied, and compared in order to forecast behaviors in markets and other areas that can affect sales and marketing managers' decisions as well as the decisions of administrators of educational institutions, hospitals, and other organizations
- **Increased security**—Although various departments may share data, access to specific information can be limited to selected users—called *authorization control*.



3 Principal Database Components



- **Data Dictionary**
 - Repository that stores the data definitions and descriptions of the structure of the data and the database
- **DBMS Utilities**
 - Programs that allow you to maintain the database by creating, editing, deleting data, records, and files
 - Also include automated backup and recovery
- **Report Generator**
 - Program for producing on-screen or printed readable documents from all or part of a database



Database Administrator (DBA)

- Coordinates all related activities and needs for an organization's database
- Ensures the database's:
 - Recoverability
 - Integrity
 - Security
 - Availability
 - Reliability
 - Performance

8.3 Database Models



A database model determines the information a database will contain and how it will be used and how the items in the database relate to one another.

Hierarchical Database

- Fields or records are arranged in related groups resembling a family tree with **child** (low-level) records subordinate to **parent** (high-level) records
- Root record is the parent record at the top of the database, and data is accessed top-down, through the hierarchy
- Oldest and simplest; used in mainframes in 1970s
- Still used in some reservation systems
- Is rigid in structure and difficult to update



Hierarchical Database

Records are arranged in related groups resembling a family tree, with “child” records subordinate to “parent” records.

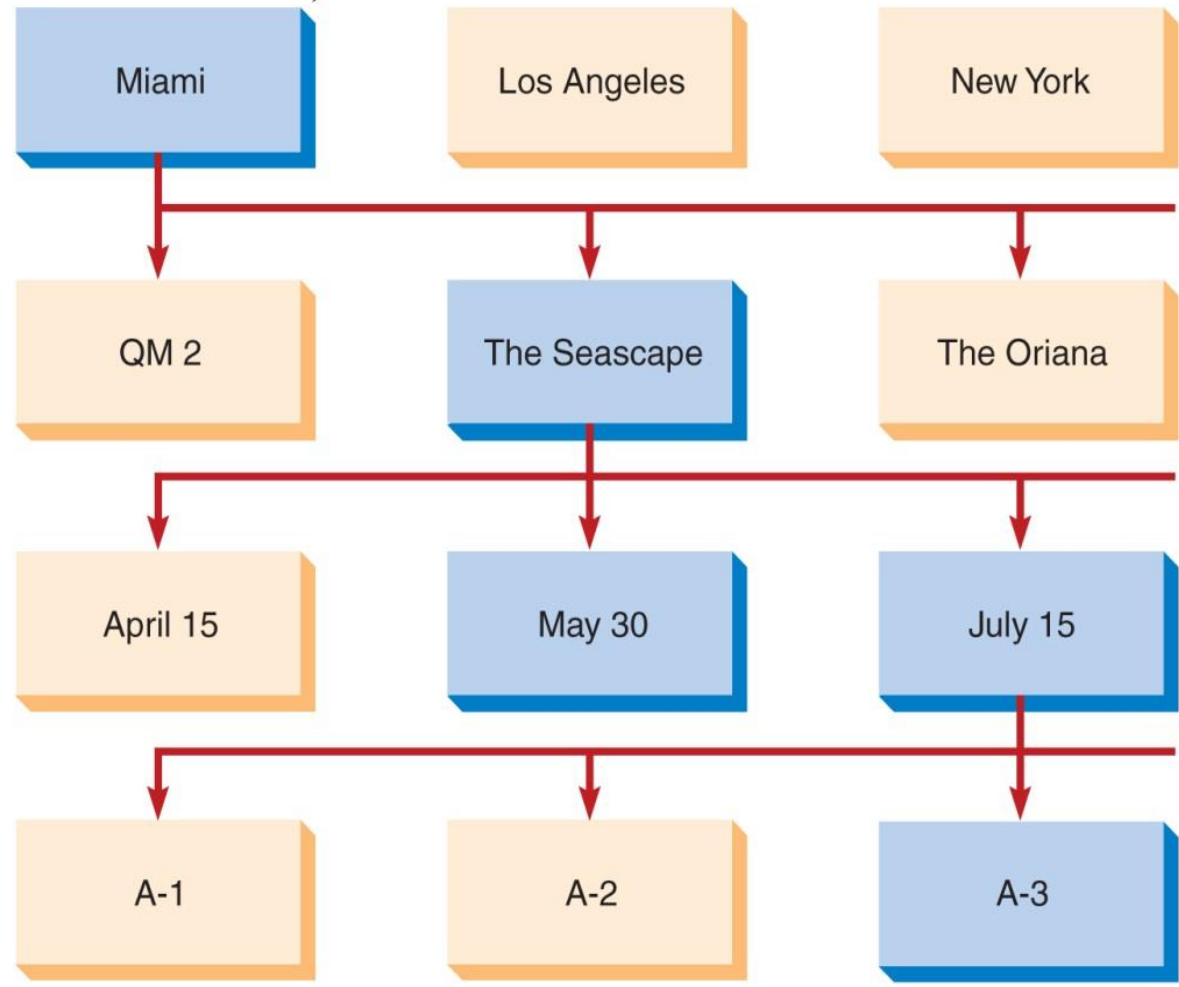
The parent at the top, Miami, is called the “root parent.”

Ports of departure

Names of ships

Sailing dates

Cabin numbers



Sailing dates (April 15, May 30, July 15) are children of the parent Seascapes.

Cabin numbers (A-1, A-2, A-3) are children of the parent July 15.



Network Database: created to represent a more complex data relationship effectively, improve database performance, and impose a database standard.

- Similar to a hierarchical database but more flexible-- each child record can have more than one parent record
- Used principally with mainframe computers
- Requires the database structure to be defined in advance; flexibility still lacking



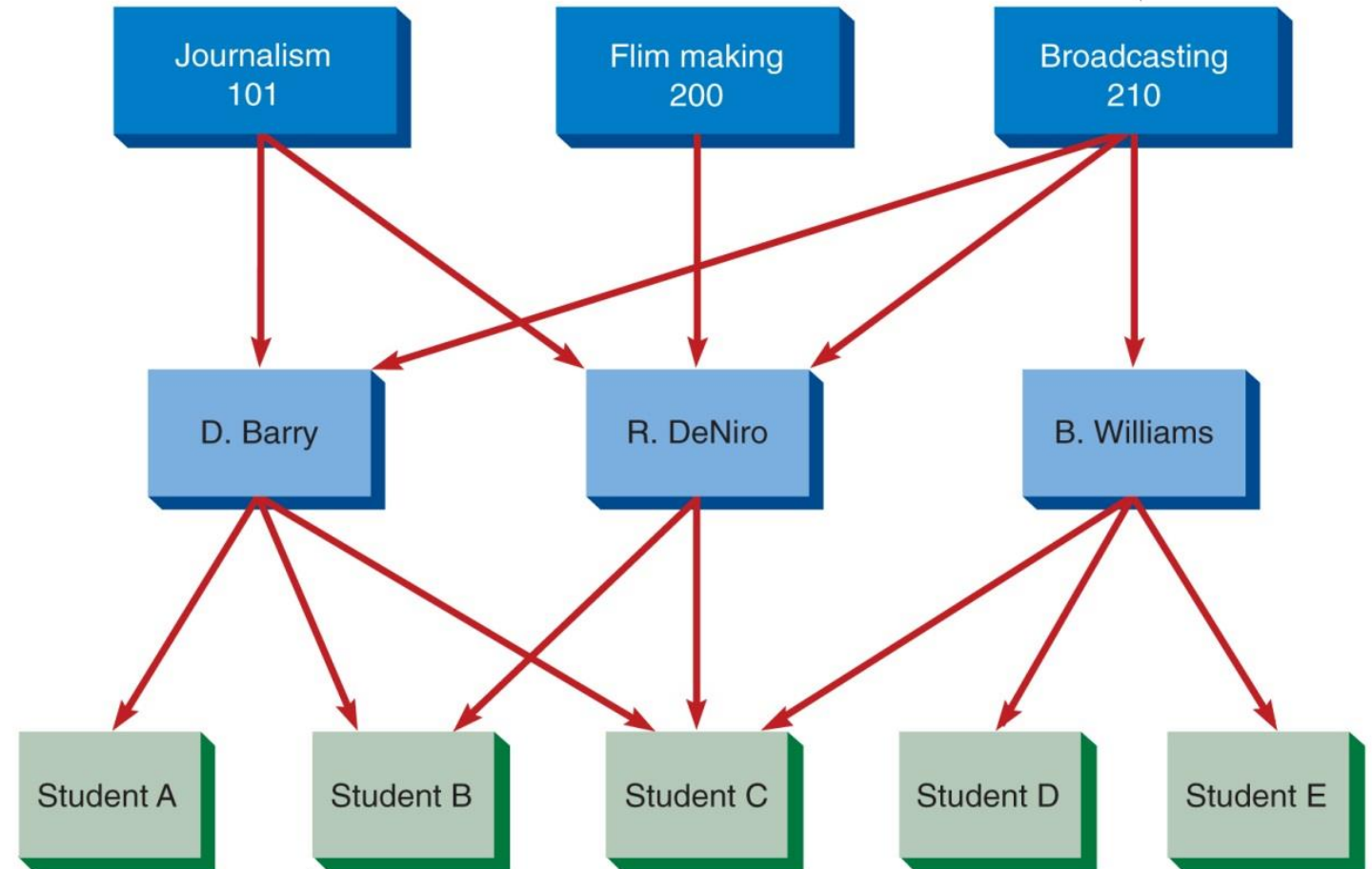
This is similar to a hierarchical database, but each child, or “member,” record can have more than one parent, or “owner.”

Courses

Network Database

Instructors

Students



The owner Broadcasting 210 has three members—D. Barry, R. DeNiro, and B. Williams.

Student B's owners are instructors D. Barry and R. DeNiro.



- **Relational Database:** grew out of the hierarchical and network database models
 - Relates or connects data in different files through the use of primary keys, or common data elements
 - Data stored in **tables** (relations, or files) of rows (tuples, or records) and columns (attributes, or fields)
 - More flexible than previous models; built with **SQL**
 - Examples for large systems are Oracle, Informix, Sybase
 - Examples for microcomputers are Paradox and Microsoft Access
 - Users don't need to know data structure to use the database



Relational Database

This kind of database relates, or connects, data in different tables (files) through the use of a key, or common data element. The relational database does not require predefined relationships.

Driver's license table

Driver's name	Street address	City	State	Zip	Driver's license number	Expiration date
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Primary keys linked

Car owner table

Car license number	Car make and year	Owner's name	Street address	City	State	Zip
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Primary keys linked

Moving violation citation table

Citation number	Moving violation type	Date cited	Driver's license number of driver cited	Fines paid/ not paid
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Primary keys linked

Parking violation citation table

Citation number	Parking violation type	Date cited	Car license number	Fines paid/ not paid
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Relational Database *(continued)*

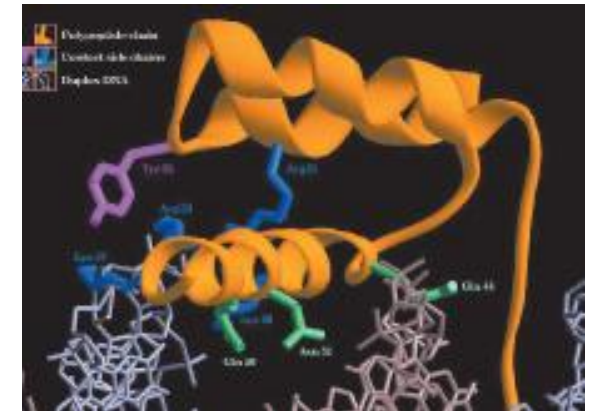
- Users employ SQL (structured query language) to create, modify, maintain, and query the database
 - **Query by Example** uses sample record forms to allow users to define the qualifications for choosing records
- Some relational database allow the use of natural spoken language to make queries

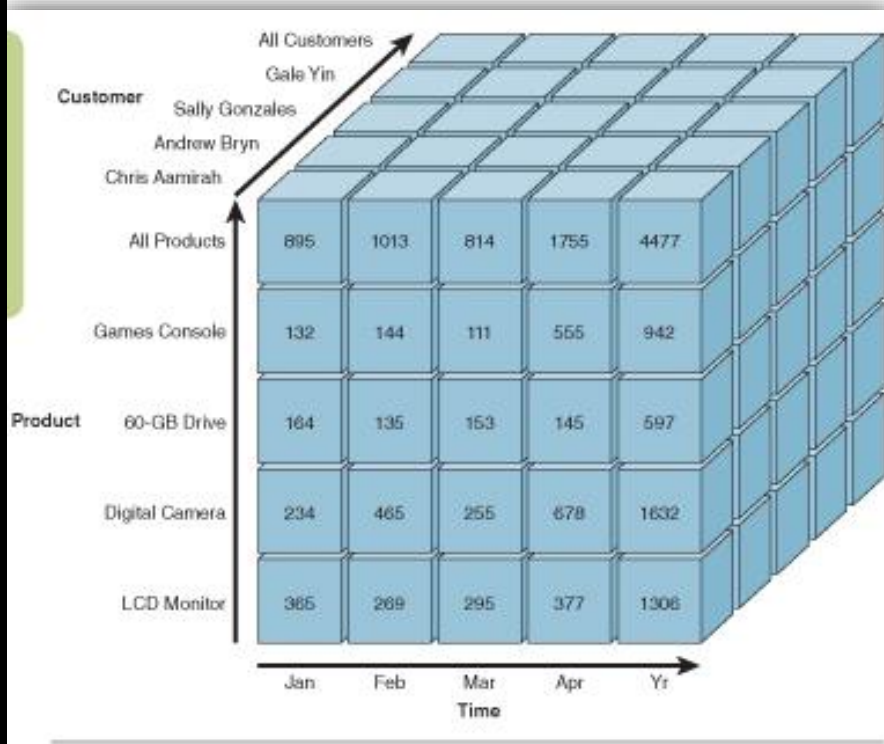




Object-Oriented Database

- Uses “objects,” software written in small, reusable chunks, as elements within data files
- An object consists of:
 - Data in any form, including audio, graphics, and video
 - Instructions on the action to be taken with the data
- This model is a **multimedia database**
- Types include web (hypertext) database and hypermedia database, which also includes links





Multidimensional Database

- Models data as facts, dimensions, or numerical answers for use in the interactive analysis of large amounts of data for decision-making purposes
- Allows users to ask questions in colloquial language
- Use OLAP (online analytical processing) software to provide answers to complex database queries



Brief Database Model Overview

Database Type	Description
Hierarchical database	Fields or records are arranged in a family tree, with child records subordinate to parent or higher-level records
Network database	Like a hierarchical database, but each child record can have more than one parent record
Relational database	Relates, or connects, data in different files (tables) through the use of a key, or common data element
Object-oriented database	Uses objects (software written in small, reusable chunks) as elements within database files; multimedia
Multidimensional database	Models data as facts, dimensions, or numerical measures for use in the interactive analysis of large amounts of data



8.4 Data Mining

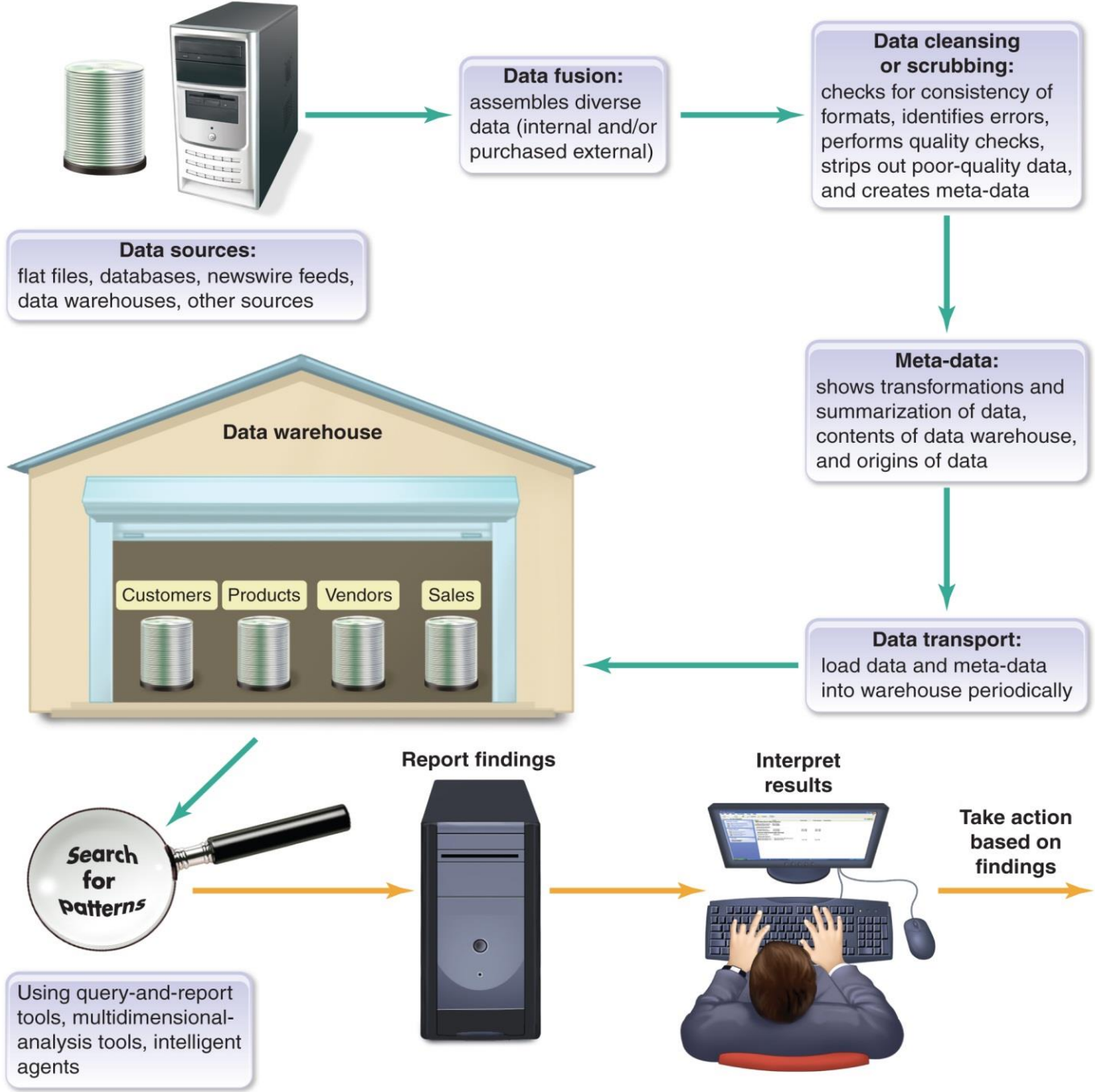


Data mining is the computer-assisted process of sifting through and analyzing vast amounts of data to extract hidden patterns and meaning and to discover new knowledge.

- Data is fed into a data warehouse through the following steps:
 1. Identify and connect to data sources
 2. Perform data fusion and data cleansing
 3. Obtain both data and metadata (data about the data)
 4. Transport data and metadata to the data warehouse
- **Data warehouse** is a special database of cleaned-up data and metadata.



Data Mining





Methods for searching for patterns in the data and interpreting the results

- **Regression analysis**
 - Develops mathematical formula to fit patterns in the data that has been extracted
 - Formula is then applied to other data sets of the same type to predict future trends
- **Classification analysis**
 - Statistical pattern-recognition process that is applied to data sets with more than just numerical data



UNIT 8B: Big Data, Information Systems, & Artificial Intelligence

Big Data aims to tap all the web data and other data that is *outside* corporate databases. Big Data typically means applying the tools of artificial intelligence to vast new sources of data beyond that captured in standard databases. The new data sources include web-browsing data trails, social network communications, sensor data, and surveillance data.



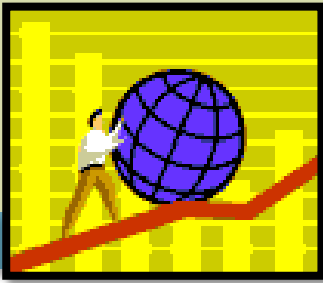
8.5 The Evolving World of Big Data



Three Implications of Big Data:

1. Big Data derives from a bundle of old & new data sources, both old and new—web pages, sensor signals, GPS location data from smartphones, browsing habits, genetic information, and surveillance videos. To make sense of the oceans of data, there is advanced computer processing and storage plus complex software taken from the evolving world of artificial intelligence, the branch of computer science devoted to the creation of computer systems that simulate human reasoning and sensation.

The software applies **Big Data analytics** – the process of examining large amounts of data of a variety of types to uncover hidden patterns, unknown correlations, and other useful information. A specific kind of analytics is *web analytics*, the measurement and analysis of Internet data to understand web usage.



2. Big data could lead to a revolution in measurement: The volume and variety of data, along with the powerful smart software, could revolutionize how things are measured—just as the invention of the telescope opened up the heavens and the microscope unveiled the mysteries of biological life down to the cellular level. In business management, for example, new kinds of measurement could replace old ideas, organizations, and ways of thinking about the world.





3. Big data could lead to better decision making: Not only can data-driven insights be used to make sense of incredibly complex situations, Big Data “can help compensate for our overconfidence in our own intuitions and can help reduce the extent to which our desires distort our perceptions.” In short, *Big Data* is a term for a process that has the potential to transform everything.



Uses of Big Data:

- Big Data is finding major uses in medical research, marketing, politics, and even entertainment programming, to name just a few areas.



8.6 Information Systems in Organizations

Using Databases to Help Make Decisions



An **information system** is a combination of people, hardware, software, communication devices, and databases that processes data and information for a specific purpose.

- What are the qualities of good information?
 - Correct and verifiable
 - Complete yet concise
 - Cost effective
 - Current
 - Accessible



Most organizations have 6 departments within which information must flow, horizontally:

- Research and development
- Production (operations)
- Marketing and sales
- Accounting and finance
- Human resources (personnel)
- Information systems (IS)



Besides the 6 departments, many organizations also have 3 levels of management, where information flows vertically:



- **Strategic-level management**
 - Top managers (CEOs, COOs, CFOs, CIOs) concerned with long-term, or strategic, planning and decisions
- **Tactical-level management**
 - Middle level managers who make tactical decisions to implement the strategic goals set for the organization
- **Operational-level management**
 - Low-level supervisors who make daily operational decisions



- A Newer Information Flow: Decentralized Organizations
 - The pyramid management structure is flattened somewhat as employees are given more authority to make day-to-day decisions.
 - Employees increasingly linked to a central database.
 - Companies use Groupware CSCW (computer-supported cooperative work) systems to enable cooperative work by groups of people.
 - Many people can work together from different locations to manage information.





Computer-based information systems: information systems that are a combination of hardware, software, and telecommunications networks that people build and use to collect, create, and distribute data.

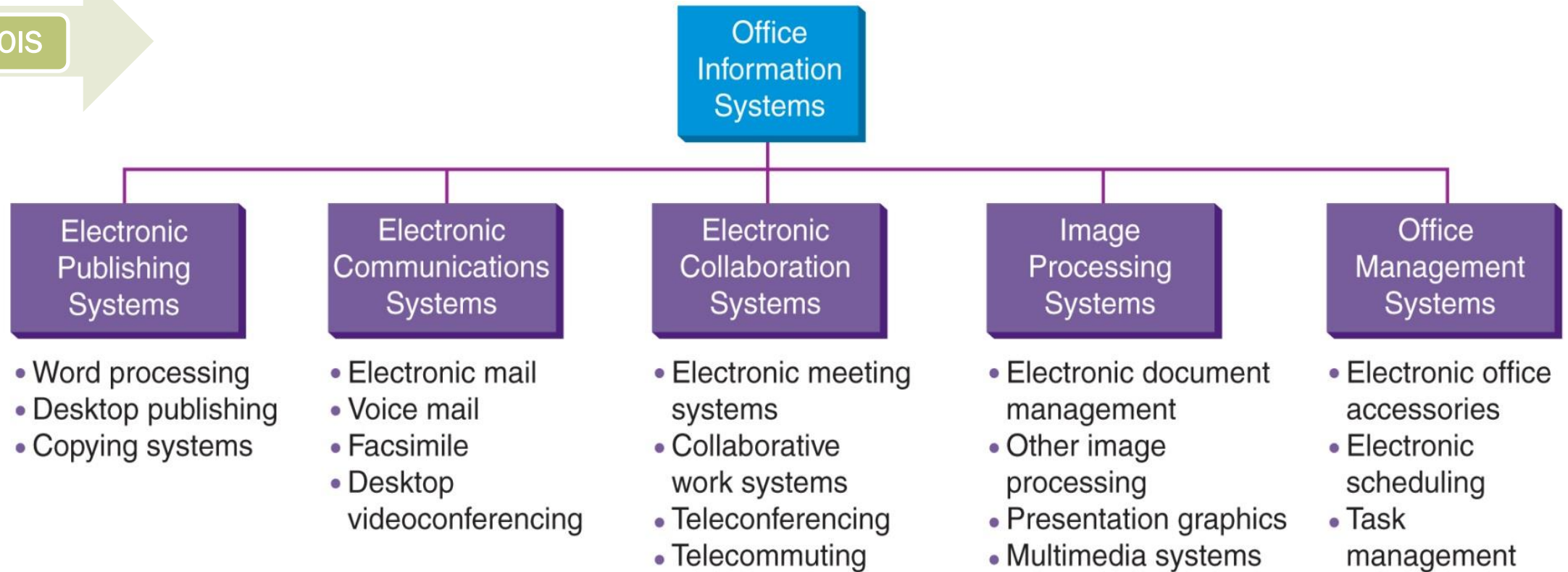
1. Office information systems
2. Transaction processing systems
3. Management information systems
4. Decision support systems
5. Executive support systems
6. Expert systems



1. Office Information System (OIS)

- Also called *office automation system*
- Combines various technologies to reduce the manual labor required in operating an efficient office and to increase productivity
- Used throughout all levels of an organization
- Uses, e.g., fax, voice mail, email, scheduling software, word processing, desktop publishing
- OIS backbone = network (LAN, intranet, extranet)

OIS





2. Transaction Processing System (TPS)

- Transactions are recorded events of routine business activities, such as bills, orders, and inventory
- TPS systems keep track of the transactions needed to conduct a business
- Features of a TPS:
 - Input and output: transaction data
 - For operational (low-level) managers
 - Produces detail reports (specific information about routine activities)
 - One TPS for each department
 - Basis for management information systems (MIS) and decision support systems (DSS)



3. Management Information System (MIS)

- Computer-based information system that uses data recorded by a TPS as input to programs that produce routine reports as output
- Features
 - Inputs are processed transaction data; outputs are summarized, structured reports
 - Designed for tactical (mid-level) managers
 - Draws from all departments
 - Produces several kinds of reports: summary, exception, periodic, and demand



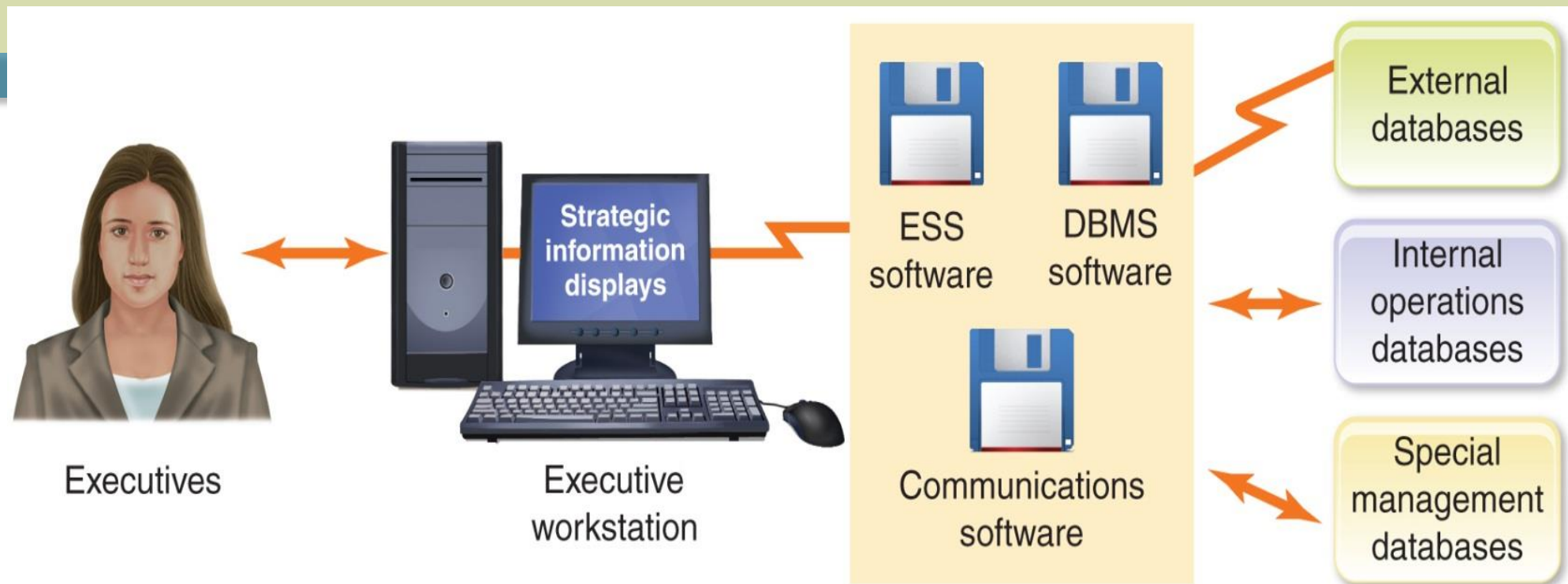
4. Decision Support System (DSS)

- Computer information system that provides a flexible tool for analysis and helps management focus on the future
- Features
 - Inputs are external data and internal data such as summarized reports and processed transaction data; outputs are demand reports from top managers
 - Assists tactical (mid-level) managers in decision making
 - Produces analytic models
- Developed to support the types of decisions faced by managers in specific industries



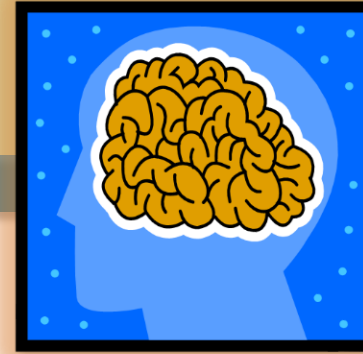
5. Executive Support System

- Easy-to-use DSS made especially for strategic (top-level) managers to support strategic decision making
- Uses data from internal systems and data from outside
- Allows executives to call up predefined reports
- Includes capability to browse through summarized information on all aspects of the organization and drill down for detailed data
- Allows executives to perform “what-if” scenarios



Executive Support System

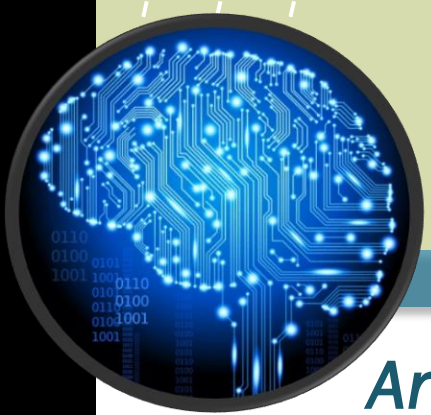
Executive support systems are reporting tools that allow organizations to turn their data into useful summarized reports. These reports are generally used by executive-level managers for quick access to reports coming from all company levels.



6. Expert System

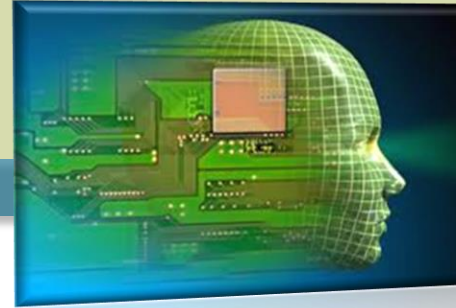
- Also called *knowledge-based system*
- Set of interactive computer programs that help users to solve problems that would otherwise require the assistance of a human expert
- Used by both management and nonmanagement personnel to solve specific problems
- One of the most useful applications of artificial intelligence (AI)

8.7 Artificial Intelligence



Artificial intelligence is the branch of computer science concerned with making computers behave like humans.

- Two approaches to AI are conventional AI, based on machine learning, and computational intelligence, based on experimental and trial-and-error methods.
 - **Conventional AI** attempts to mimic human intelligence through logic and symbol manipulation, as well as statistics. This branch of AI is based on *machine learning*, which is the development of techniques that allow a computer to simulate learning by generating rules from raw data fed into it. Expert systems, for example, make heavy use of this kind of AI.
 - **Computational intelligence** relies less on formal logical systems and more on experimental and trial-and-error methods. This branch of AI is based on heuristics, or rules of thumb, for solving a problem, rather than hard-and-fast formulas or algorithms.



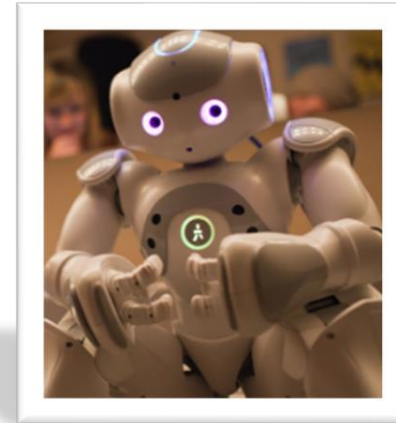
Weak AI versus Strong AI:

- **Weak AI** makes the claim that computers can be programmed to *simulate* human cognition and only *some* human cognition, to solve particular problems or reasoning tasks that do not encompass fully human intelligence. That is, weak AI suggests that some “thinking-like” features can be added to computers to make them more useful tools.
- **Strong AI** makes the claim that computers can be made to think on a level that is at least equal to humans and possibly even be conscious of themselves. So far, most AI advances have been piecemeal and single purpose, such as factory robots. However, proponents of strong AI believe that it’s possible for computers to have the kind of wide-ranging problem-solving ability that people have.



AI Areas include:

- Expert systems
- Natural language processing
- Intelligent agents
- Pattern recognition
- Virtual reality and simulation devices
- Robotics
- Fuzzy logic
- Neural networks



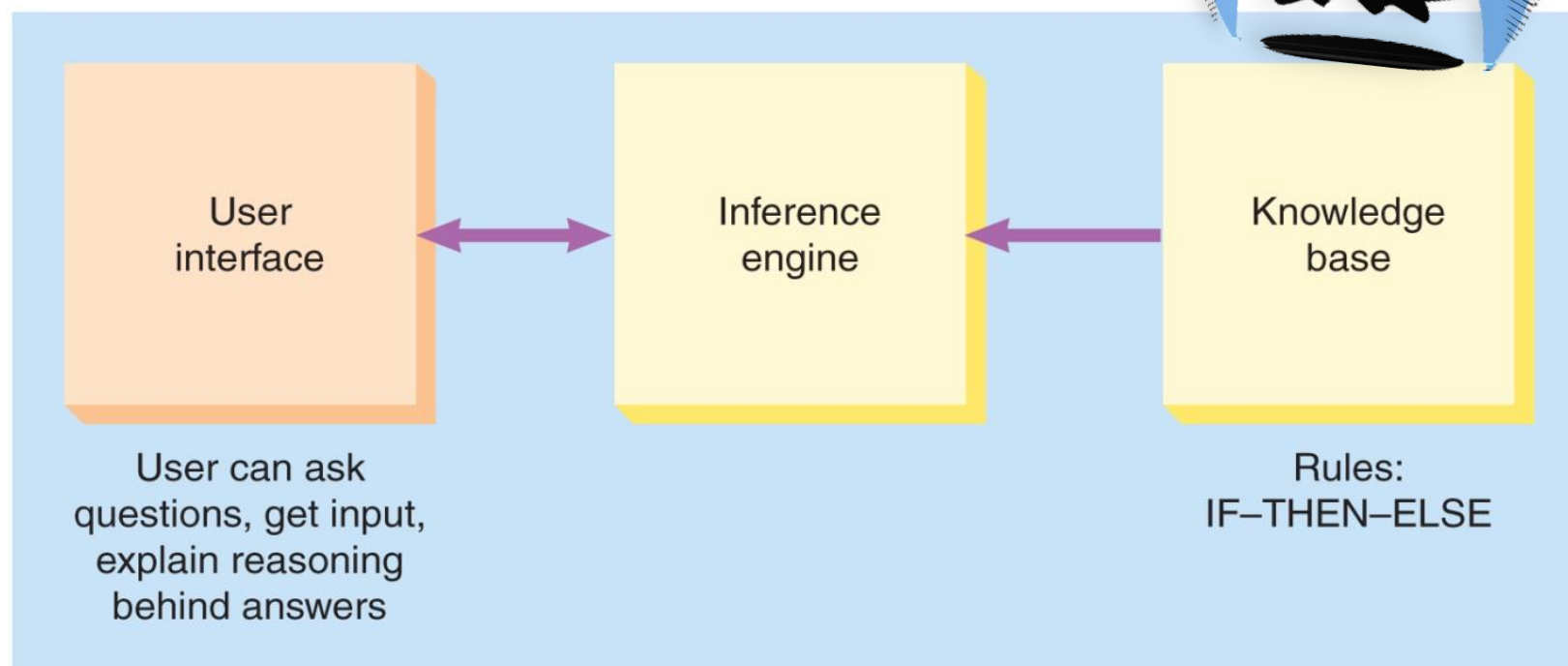


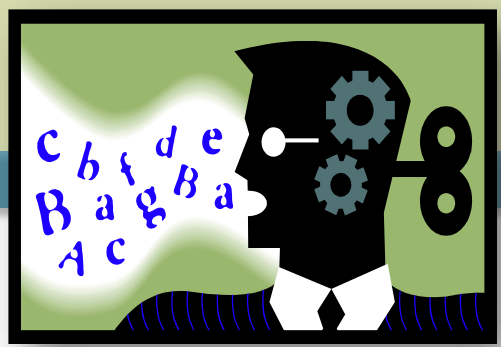
Expert Systems

- Built by knowledge engineers
- Include surface knowledge and deep knowledge
- Three components of an expert system:
 - **Knowledge base:** an expert system's database of knowledge about a particular subject
 - **Inference engine:** the software that controls the search of the expert system's knowledge base and produces conclusions
 - **User interface:** the display screen for the user to interact with the expert system



Expert System





Natural language processing

- Allows users to interact with a system using normal language
- The study of ways for computers to recognize and understand human language

Intelligent agents



- A form of software with built-in intelligence that monitors work patterns, asks questions, and performs work tasks on your behalf; shop bots are intelligent agents

Pattern recognition

- Involves a camera and software that identify recurring visual patterns by mapping them against similar patterns stored in a database (e.g., visual surveillance and ID of suspicious people)

Virtual reality & simulation devices



- A computer-generated artificial reality that projects a person into a sensation of 3-D space
- Often used as simulators to represent the behavior of physical or abstract systems—e.g., for pilot training

Robotics

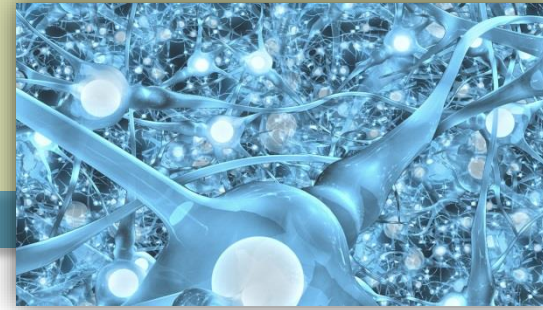
- The development and study of machines that can perform actions that are normally done by people
 - **Robots grouped by locomotion system:** grouped according to their means of locomotion, which defines their shape. Thus, there are stationary, wheeled, legged, swimming, flying, rolling, swarm, modular, micro, nano, soft elastic, snake, and crawler robots (includes drones).
 - **Robots grouped by application:** grouped according to the application they are supposed to perform, so that shape is not important. Thus, in health and medicine, there are wearable machines to help amputees walk, wheeled robots (medi-bots) that roam hospital halls and make visits to patients on behalf of their doctors, and robots used in surgery that perform actual operations.



Fuzzy logic

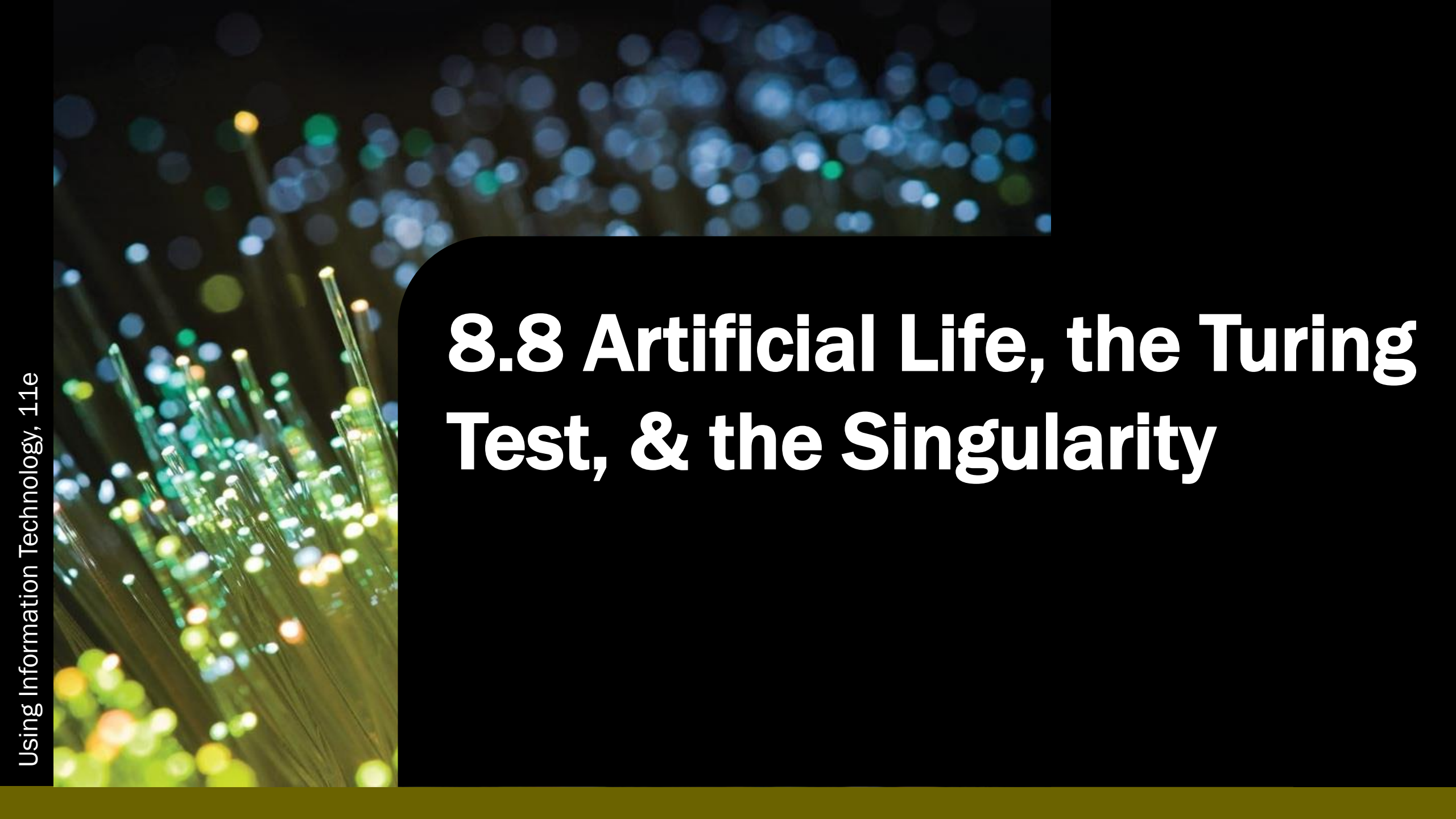
- A method of dealing with imprecise data and uncertainty, with problems that have many answers rather than one
- Similar to human logic
- Has been applied in running elevators to determine optimum times for elevators to wait; used in many appliances





Neural network: consists of a network of processors that are interconnected in a way that is similar to the connections between neurons, or nerve cells, in the human body. The neural network is able to simulate the behavior of biological neural networks, as in pattern recognition, language processing, and problem solving. A neural network is able to learn from example and does not require detailed instructions.

- Neural networks have been used in machine vision, credit-card fraud detection, and diagnosis of heart attacks.



8.8 Artificial Life, the Turing Test, & the Singularity



Artificial intelligence leads to the question of how can we know a machine is truly intelligent, which figures in the Turing test.

Turing Test: In 1950 Allen Turing predicted computers would eventually be able to mimic human thinking.

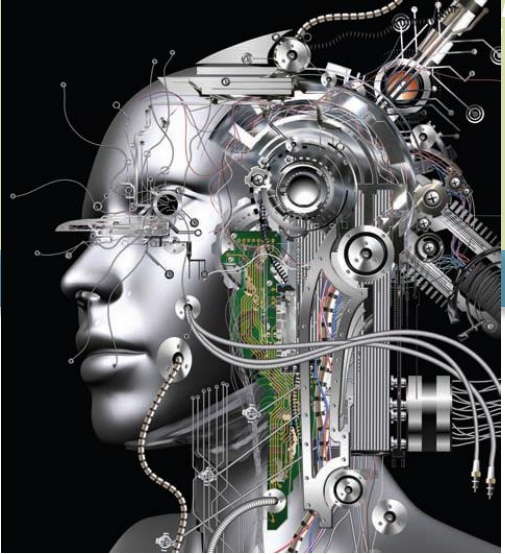
- Turing test determines whether the computer is human
 - Judge is in another location and doesn't see the computer
 - Judge converses via a computer terminal with two entities: one a person and one a computer
 - Judge must determine who is the person and who is the computer
 - If the computer can fool the judge, it is said to be intelligent
 - No computer system has yet passed the Turing test



Smarter-Than-Human Computers

The Singularity

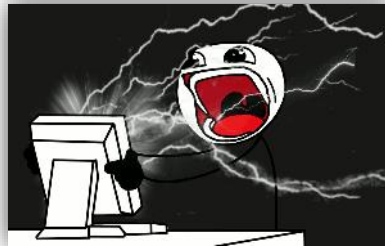
- A moment when humans would have created self-aware, smarter-than-human machines capable of designing computers and robots that are better than humans can design today
- Also may involve transferring the contents of human brains and thought processes into a computing environment

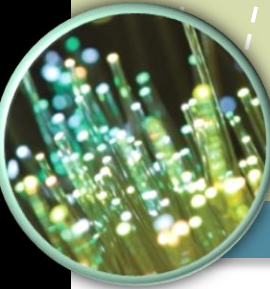




Ethics in A.I.

- Ethics underlies everything having to do with AI.
- Computer software is subtly shaped by the ethical judgments and assumptions of its creators; there is no human-values-free / bias-free software.
- Will AI cause humans to lose control of computer systems?





Databases: Concerns about Privacy & Identity Theft

- Databases have facilitated loss of privacy and identity theft, which have become significant concerns for many people.

